SynapSign: An Advanced Machine Learning Framework for American Sign Language Recognition Utilizing a Novel Landmark-Based Dataset

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***Abstract*—Hearing loss is a common condition affecting a significant proportion of the world's population, creating barriers to effective communication. Sign language, particularly American Sign Language (ASL), is an important tool for the social integration and personal growth of people with hearing loss. The need for effective tools to facilitate the learning and practice of ASL is increasingly recognized. Although there are many studies and proposed softwares that execute based on certain approaches such as recognition of signs by machine learning techniques with relatively high accuracy, it is apparent that there exists a need for higher performance. This paper presents SynapSign, a desktop application designed to enhance ASL learning using machine learning algorithms. To develop the application using the machine learning model having the highest accuracy, the performances of Random Forest, XGBoost and Deep Neural Network (DNN) classifiers were investigated. To this purpose, an image dataset consisting of 2600 images was prepared. For each letter of 26 letters for ASL, 100 images of hands showing 21 hand landmarks was built for accurate hand gesture recognition using Google's MediaPipe technology. Thereafter, the three classifiers were trained on this extensive dataset of ASL hand images. Acquired models were tested and their performances were compared based on accuracy, precision and recall metrics. The results reveal that the model of Random Forest classifier performs slightly higher for all three metrics with 99.6%, 99.3% and %99.7, respectively, than other models. Therefore, SynapSign was developed using this model with a user interface that allows user to input sign image from a video stream via a camera and labelled with 21 hand landmarks by MediaPipe Framework's default hand detection model. Compared to traditional methods, the application provides a more interactive and engaging learning experience, allowing users to practice and improve their ASL skills with real-time feedback. Our findings suggest that SynapSign could serve as a valuable tool for both educational and accessibility purposes, addressing the gap in resources available to ASL learners.**

Keywords—Hearing Impairment, Interaction System, Sign Language, Random Forest, XGBoost, Deep Neural Network (DNN), Hand Gesture Recognition, Machine Learning

I. INTRODUCTION

The World Health Organization (WHO) estimates that approximately 466 million individuals worldwide are affected by hearing impairment, a condition that poses significant challenges to daily communication [1]. Hearing impairment, as defined by the WHO, is a condition that severely hinders the ability to understand speech, for which sign language has proven to be a highly effective solution [2]. Therefore, sign language is a vital communication tool for the deaf community, however, it serves not only as a communication tool for the hearing impaired but also as a crucial instrument for social integration and personal development. There exist types of sign language with respect to spoken languages, such as American Sign Language (ASL).

Numerous studies have been conducted on teaching, learning and practicing sign language. Learning sign language is reported to enhance cognitive flexibility and language acquisition, while technology-supported learning tools further facilitate the process and increase motivation [3]. The cognitive and social benefits of learning ASL have garnered significant attention. As a result, the ASL alphabet is chosen as the primary mode of communication, with research underscoring the importance of phonological features, such as hand shape, in sign production [4]. ASL is now recognized as the third most common language in the United States, gaining prominence due to its deep roots in Deaf culture [1].

The advent of mobile technology has spurred the development of numerous language-learning applications. However, a significant lack of means and resources exists for those wishing to learn sign language. To address this gap, our research group has undertaken the development of an automatic sign language recognition system. Developing an automatic sign language recognition system presents several challenges. Nonetheless, several models targeting recognition of ASL have been constructed using the MediaPipe Framework and Long Short-Term Memory (LSTM)

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algorithms, particularly for detecting gesture-based ASL expressions, as illustrated in “Fig. 1” [1]. These efforts aim to facilitate communication within the deaf and hearing-impaired community [5].

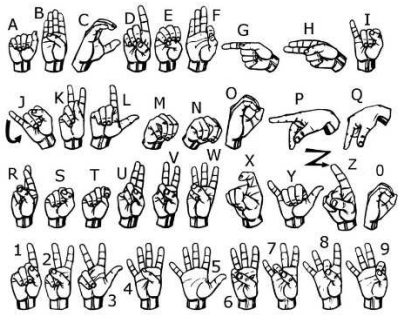


Fig. 1. American sign language

The effectiveness of immersive learning environments for ASL learners has been demonstrated, with findings showing that such environments enhance both receptive and expressive skills [6]. The impact of mobile applications on ASL learners has also been examined, revealing that these tools significantly enhance vocabulary and signing accuracy [7]. The use of virtual reality (VR) technology in sign language education has been explored, with findings showing that VR increases learner interest and motivation, thereby improving skill retention [8]. The impact of online courses has been examined, with results indicating that online learners achieve similar or even superior progress compared to traditional classroom learners [9]. The advantages of community engagement for sign language learners have been demonstrated, with those engaging regularly showing increased confidence and competence [10]. A 2021 study examined the impact of sign language clubs on learner motivation and skill development, finding that participants in these clubs experienced significant improvements in both motivation and signing abilities [11]. However, current methods for teaching and practicing ASL face several limitations. Firstly, there is a notable lack of standardized assessment tools to measure ASL abilities among deaf learners, leading practitioners to rely on modified English-based assessments [12]. Secondly, the scarcity of large-scale datasets related to ASL comprehension and production hampers the advancement of deep learning research in sign languages. Moreover, the existing datasets are often inadequate for practical application [13]. Finally, despite some progress in developing ASL training applications, the absence of real-time sign recognition systems significantly limits the effectiveness of these learning tools [14]. Therefore, there is a clear need for the development of more diverse and accurately labeled datasets [15]. These studies collectively demonstrate the effectiveness of various methodologies and technological innovations in sign language teaching and practice, providing learners with diverse options to achieve greater proficiency and confidence. The objective of this study is to enhance opportunities for students to practice sign language learning. To this end, we developed an application using a model trained on our own dataset.

A. Literature Review

Research has shown that the use of virtual reality (VR) technology in sign language education increases learner interest and motivation, leading to better retention of skills [8]. In the literature, MediaPipe's real-time hand gesture recognition capabilities have been successfully applied to various virtual reality applications and have been used efficiently in human-computer interactions [26]. Taken together, these results highlight the effectiveness and versatility of MediaPipe in enhancing gesture recognition systems in various applications and scenarios. MediaPipe's high success rates in hand gesture recognition are highlighted throughout the literature, and similar results were obtained in this study. This study extends MediaPipe's success in recognising ASL hand gestures and improves its accuracy by combining it with algorithms such as Decision Tree and DNN.

II. Materials and Methods

The objective of this study is to develop a software application with a machine learning model that is capable of identifying the letters of the American Sign Language (ASL) alphabet with high accuracy using image data. The project employs sophisticated techniques, including Random Forest, XGBoost, and Deep Neural Networks (DNN), to harness the capabilities of MediaPipe for precise hand landmark extraction and gesture recognition. The incorporation of these methodologies into the SynapSign application represents a novel approach to enhancing real-time sign language recognition, with a focus on accuracy, usability, and accessibility.

A. Random Forest Classifier

The decision tree is selected for its simplicity and interpretability, with the data space being partitioned iteratively and models constructed in each partition. A random forest classifier is a powerful ensemble learning method that consists of multiple decision trees trained on random subsets of the data. It addresses issues like block under-utilization and data over-read in training algorithms [16]. Additionally, random forest classification proved to be highly effective in healthcare, specifically in predicting individual responses to transcranial direct current stimulation for tinnitus relief, outperforming traditional statistical approaches with high accuracy and identifying key predictors of treatment response [17].

B. XGBoost (Extreme Gradient Boosting)

XGBoost is a powerful machine learning algorithm commonly used for hand gesture classification tasks. Research studies have shown that XGBoost, along with other classifiers like Random Forest, Support Vector Machine (SVM), and Convolutional Neural Networks (CNN), has been implemented successfully in classifying hand gestures with high accuracy [18]. XGBoost has demonstrated impressive accuracy rates, with studies reporting values close to 97% in hand gesture recognition tasks, making it a popular choice among researchers for its efficiency and effectiveness in classification tasks [19].

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C. Deep Neural Network (DNN)

Deep neural network (DNN) classifiers differentiate hand gestures by leveraging advanced techniques such as automatic feature extraction and deep learning models. Various studies have shown the effectiveness of DNNs, including convolutional neural networks (CNNs) and multistream fusion 1-D CNNs, in accurately classifying hand gestures [20] [21] [22] [23]. These models excel in capturing intricate details like hand shape, motion profiles, and positioning, crucial for accurate gesture identification [23].

D. Modality And Articulations

A unimodal approach was used, as the system in question utilizes images captured by the camera as input and attempts to comprehend sign language by processing hand gestures within these images. In this context, the user is required to perform the following tasks:

- Hand Gestures and Signs: The user will be required to perform hand gestures in order to create certain signs.
- Finger Positions: The position of the fingers and the hand as a whole can be used to determine the meaning of a sign.
- Hand Coordination: The coordination of hand movements is of significant importance for the accurate transmission of signs.

E. Dataset

A machine learning model was developed specifically to identify the letters of the American Sign Language alphabet. A dataset of 2600 2D images was built, consisting of 100 images with 1280x720 resolution and 16:9 aspect ratio for each letter of ASL alphabet (i.e., 26 letters). In addition, there are 21 landmarks for each hand image as shown in "Fig. 2". The data collection process was carried out through the camera over the indices allocated for each class and this process was repeated for each class. Hand gestures were identified, coordinates extracted and normalized from hand images using the MediaPipe hands module. These data and labels were saved in a pickle file for use in training machine learning models.

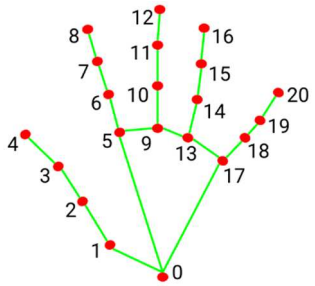


Fig. 2. The analytical localization of 21 hand landmarks.

F. Development Approach and Tools

American Sign Language (ASL) hand gestures are identified using MediaPipe technology with Decision Tree and Deep Neural Network (DNN) classifiers. Previous studies have shown that the MediaPipe technology achieves very high accuracy rates in recognising ASL hand gestures, and that the feature extraction process performed on this data yields successful results [24][25]. Additionally, in the context of virtual reality technology, Mediapipe, in conjunction with OpenCV, is employed for real-time hand gesture recognition, enabling efficient human-computer interaction in virtual environments. A multilayer network is trained on the hand sign data using the DNN method, which models complex patterns through non-linear relationships, employing activation functions like ReLU to capture intricate data patterns. ReLU enhances the expressivity of neural networks, allowing them to approximate any continuous function with high precision when the network is wide enough [27]. MediaPipe also assists in the identification and tracking of hand gestures, extracting key landmarks that are used for training both classifiers. This dual-method approach utilizes the simplicity of decision trees and the deep learning capabilities of DNNs, enhancing robustness in hand gesture recognition.

G. System Design

System design is a process that defines how the components in an often-complex engineering project will fit together and how they will work together [28]. This includes the process of specifying the functions and performance designed to meet the requirements of a product, system or process [28].

The study is based on users initiating the process to detect sign language gestures through their cameras. The images captured by the camera are subjected to direct analysis by a pre-trained model, which is then used to recognize specific sign language gestures. Based on the outcome of the analysis, the model determines whether the gesture has been correctly identified. Whenever a sign is recognized, the system displays the translation result to the user. This process is performed using advanced object detection and classification algorithms as MediaPipe and deep learning models.

The Sign Language Recognition System incorporates integration with the camera device, enabling users to log in and manage their user account with a range of functions shown in Figure 3. This integrated system facilitates communication for individuals with hearing impairments by effectively performing sign language recognition and translation.

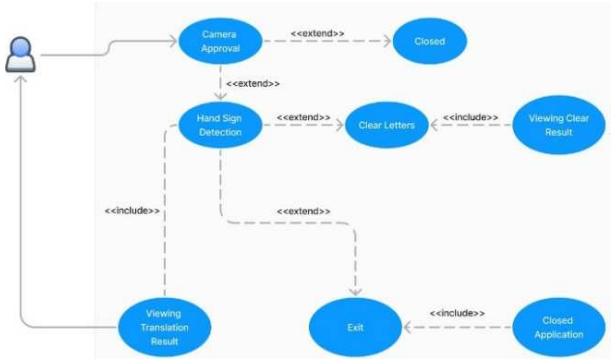


Fig. 3. Use case diagram

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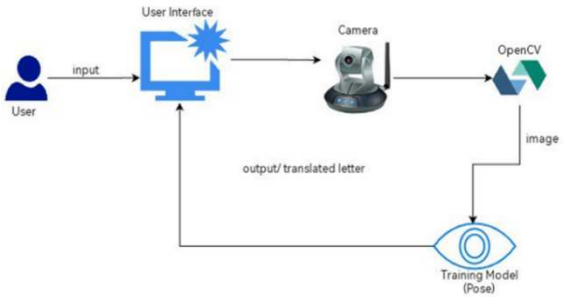


Fig. 4. System architecture

H. Interface

The SynapSign application, which was developed as part of the project, was designed to facilitate the demonstration of letters in sign language. Upon initial use of the app, users are prompted to grant access to the camera. Once permission has been granted, the application login screen appears, as shown in Figure 5. The camera is then activated and users can point their hands at the camera to display the letters in sign language. The application will display the letter corresponding to the character shown, and these letters will continue to appear as text on the screen until the application is closed or the Clear button is pressed, as shown in Figure 6 .



Fig. 5. Interface splash screen

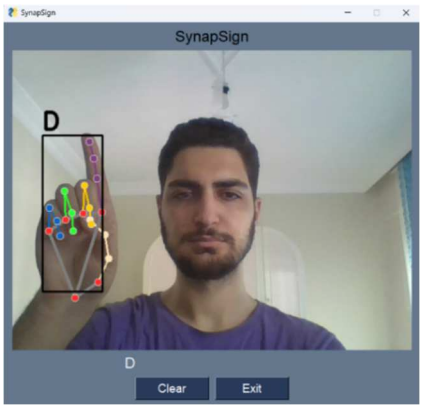


Fig. 6. User interface letter detection

III. Results And Discussion

A. Results

The Random Forest model exhibited the highest performance, with a recall of 0.997, an accuracy of 0.996, a precision of 0.993. In contrast, the XGBoost model exhibited slightly lower values, achieving 0.993 recall, 0.992 accuracy, 0.985 precision. In contrast, the DNN model exhibited a comparatively inferior performance, achieving a recall rate of 0.974, an accuracy rate of 0.984, a precision rate of 0.986. The results demonstrate that the Random Forest model is the most effective model, exhibiting particularly high accuracy and recall rates.

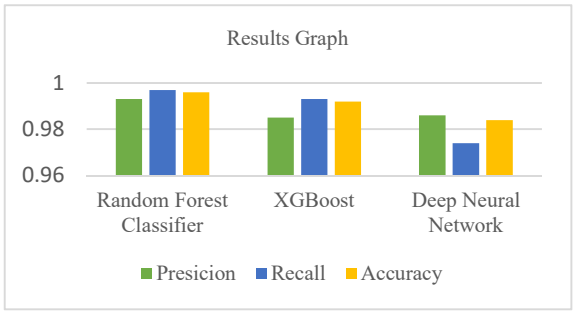


Fig. 7. Results graph

B. Discussion

It is conceivable that the project could be further expanded by incorporating additional sign languages in the future. Furthermore, augmenting the dataset with images of individuals representing diverse demographic groups will enhance the model's precision. In subsequent stages, in addition to the letters presented in fixed images, the detection of letters formed as a result of movement (e.g., the letters "j" and "z") and phrases will also be incorporated, allowing the project to advance. Subsequently, the desktop application developed for the project can be enhanced to enhance its user-friendliness and compatibility with different platforms.

The study evaluated the performance of Random Forest, XGBoost and Deep Neural Network (DNN) models using Precision, Recall and Accuracy metrics. The Random Forest model showed the highest performance with 99.3% Precision, 99.7% Recall and 99.6% Accuracy. These results show that Random Forest provides high accuracy and reliability, but previous studies have not achieved this performance with results such as 77.94% in [29], 87.1% in [30], and 96.3% in [31]. The XGBoost model provides 98.5% Precision, 99.3% Recall and 99.2% Accuracy, providing high Recall and high Accuracy, but slightly behind Random Forest.

The DNN model performed the lowest with 97.4% Precision, 98.6% Recall and 98.4% Accuracy. This result shows that although DNN has high Recall, it lags behind Random Forest and XGBoost in Precision. Comparisons show that Random Forest is the most effective model in this context and XGBoost offers a competitive alternative. The lower Precision of DNN may be limiting for certain applications.

TABLE I. ML SCORE METRICS

Model	Precision	Recall	Accuracy
Random Forest	**99.3%**	**99.7%**	**99.6%**
XGBoost	98.5%	99.3%	99.2%
DNN	97.4%	98.6%	98.4%
Random Forest [29]	-	-	77.94%
Decision Tree [30]	87.2%	87.1%	87.1%
Random Forest [31]	96%	96%	96.3%

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In this study, the performance of different machine learning models was evaluated with metrics including accuracy, precision and recall. Nevertheless, these performance metrics may not be sufficient for a final prototype. Particularly in real-time applications, it is necessary to consider the response time of the models and decide which model is more appropriate to implement. Therefore, in future studies, it is crucial to examine the response time in model selection and include it in the decision-making process.

IV. Conclusion

To observe the development and effectiveness of SynapSign, a desktop application designed to aid in the learning and practice of American Sign Language (ASL). The application employs machine learning models, including Random Forest, XGBoost, and DNN, to effectively recognize American Sign Language (ASL) hand gestures and provide users with an interactive and educational platform. In particular, the Random Forest model showed higher performance in terms of recall, accuracy and precision. In addition, it was also more accurate in practical use with the camera. Future enhancements to the application, including the incorporation of dynamic sign recognition and the expansion of the dataset, will enhance its efficacy and usability. SynapSign has the potential to facilitate improved communication, reduce feelings of isolation, and enhance the quality of life of individuals with hearing impairments.

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